

Algebra II with Trigonometry

Chapter 6.3

Mental Math (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4531393_alg-2trig-h-chp-6-3-note.pdf](#)

Standard Period What is the standard period of the base sine and cosine functions before any transformations?

Finding Amplitude What is the amplitude of the function $y = -3 \sin(x)$?

Graphing Patterns According to the notes, what is the five-point sequence (using the words High, Middle, and Low) for one complete cycle of a positive cosine wave ($+\cos t$)?

Starting Point Based on the five-point sinusoidal patterns, does a positive sine graph ($+\sin t$) begin its cycle at its high point, low point, or middle point?

Calculating Period What is the period of the function $y = \sin(4x)$?

Stretching vs. Shrinking Does the equation $y = \cos(5x)$ represent a horizontal stretch or a horizontal shrink compared to the standard cosine graph?

Vertical Shift What is the vertical shift of the function $y = -2 + \cos(x)$?

Horizontal Shift In which direction and by what value does the graph of $y = \sin(x - \pi)$ shift horizontally?

Maximum Output By combining the vertical shift and the amplitude, what is the highest possible y -value reached by the function $y = 5 + 2 \sin(x)$?

Application Period A tuning fork's vibration is modeled by the function $v(t) = 0.7 \sin(880\pi t)$. What is the period of this vibration?